



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

❑ Summary of methodologies

- Data Collection using API
- Data Collection using Web Scraping
- Data Wrangling
- Exploratory Data Analysis with SQL
- Exploratory Data Analysis with Data Visualization
- Interactive Visual Analysis using Folium
- Machine Learning Prediction

❑ Summary of all results

- Exploratory Data Analysis Visualization Results
- Interactive Analytics Visualization Results
- Machine Learning Prediction Results

Introduction

Background And Context of Project

SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage. Therefore if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against SpaceX for a rocket launch. The goal of the project is to create a machine learning pipeline to predict if landing of Falcon 9 First Stage will be successful.

Business Problems to find answered:

- What factors determine whether the Falcon 9 rocket will land successfully?
- How do different launch features interact to influence landing success rates?
- Which operational conditions are necessary to ensure a successful landing program?
- Which machine learning model performs best for predicting first-stage landing success?
- How can predictive insights support strategic decision-making in the commercial aerospace industry?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data was collected using SpaceX API and Web Scraping Wikipedia.
- Perform data wrangling
 - One-hot Encoding was applied to categorical features
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Multiple classification models were built and evaluated to predict Falcon 9 first-stage landing success.

Data Collection

Data was collected using multiple data collection techniques:

- Launch data was retrieved through GET requests to the **SpaceX API**.
- The API response was decoded into JSON format using the `.json()` function and converted into a pandas dataframe using `.json_normalize()`.
- Data cleaning and preprocessing were performed by handling missing values and correcting inconsistencies where necessary.
- Web scraping techniques were also applied to extract Falcon 9 launch records from Wikipedia using **BeautifulSoup**.
- HTML tables containing launch records were parsed and transformed into pandas dataframes for further analysis.
- The processed dataset was then used for exploratory analysis, visualization, and predictive modeling.

Data Collection – SpaceX API

- We used the get request to the SpaceX API to collect data, clean the requested data and did some basic data wrangling and formatting.
- Here is the link to the notebook <https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/Data%20Collection%20Using%20API.ipynb>

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
[10]: static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'
```

We should see that the request was successful with the 200 status response code

```
[11]: response.status_code
```

```
[11]: 200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
[12]: # Use json_normalize method to convert the json result into a dataframe
response.json()
data = pd.json_normalize(response.json())
```

Using the dataframe `data` print the first 5 rows

```
[13]: # Get the head of the dataframe
data.head()
```

```
[13]: static_fire_date_utc  static_fire_date_unix  net  window  rocket  success  failures  details  crew  ships  capsules  payload
0      2006-03-17T00:00:00.000Z      1.142554e+09  False  0.0  5e9d0d95eda69955f709d1eb  False  [{"time": 33, 'altitude': None, 'reason': 'merlin engine failure'}]  Engine failure at 33 seconds and loss of vehicle  []  []  []  [5eb0e4b5b6c3bb0006eeb1
```


Data Collection - Scraping

- Web scraping techniques were applied to extract Falcon 9 launch records from Wikipedia using BeautifulSoup.
- The extracted HTML tables were parsed and converted into pandas dataframes for further analysis and preprocessing.
- The link of the notebook: <https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/Data%20Collection%20Using%20Web%20Scraping.ipynb>

```
TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

[15]: # use requests.get() method with the provided static_url and headers
      # assign the response to a object
      response = requests.get(static_url, headers = headers)

[16]: print(response.content)

r, November, December", "wgRequestId": "9f581204-ea9b-40a5-8091-939eab907401", "wgCanonicalNamespace": "", "wgCanonicalSpecialPageName": false, "wgNamespaces": {"wgPageName": "List of Falcon 9 and Falcon Heavy launches", "wgTitle": "List of Falcon 9 and Falcon Heavy launches", "wgCurRevisionId": 1357573395, "wgRevisionId": 1027686922, "wgArticleId": 37574004, "wgIsArticle": true, "wgIsRedirect": false, "wgAction": "view", "wgUserName": null, "wgUserGroups": ["**"], "wgCategories": ["Source attribution", "CS1 maint: multiple names: authors list", "All articles with dead external links", "Articles with dead external links from February 2021", "Articles with permanently dead external links", "CS1 Spanish-language sources (es)", "CS1 Indonesian-language sources (id)", "CS1 errors: generic name", "CS1 maint: numeric names: authors list", "CS1 errors: external links", "CS1 maint: url-status", "CS1 German-language sources (de)", "CS1 Korean-language sources (ko)", "CS1 maint: deprecated archival service", "Articles with short description", "Short description is different from Wikidata", "Use American English from January 2021", "All Wikipedia articles written in American English", "Use dmy dates from January 2021", "Articles containing potentially dated statements from March 2018", "All articles containing potentially dated statements", "All articles with failed verification", "Articles with failed verification from May 2021", "Articles containing potentially dated statements from April 2020", "Pages using multiple image with auto scaled images", "Featured lists", "Articles containing video clips", "Falcon 9 and Falcon Heavy launches", "SpaceX launch vehicles", "Lists of rocket launches"], "wgPageViewLanguage": "en", "wgPageContentLanguage": "en", "wgPageContentModel": "wikitext", "wgRelevantPageName": "List of Falcon 9 and Falcon Heavy launches", "wgRelevantArticleId": 37574004, "wgTempUserName": null, "wgIsProbablyEditable": true, "wgRelevantPageIsProbablyEditable": true, "wgRestrictionEdit": [], "wgRestrictionMove": [], "wgNoticeProject": "wikipedia", "wgFlaggedRevsParams": {"tags": {"status": {"levels": 1}}, "wgConfirmEditCaptchaNeededForGenericEdit": "hcaptcha", "wgConfirmEditForceShowCaptcha": false, "wgConfirmEditCaptchaVisualEditorOnLoadIntegrationEnabled": true, "wgConfirmEditCaptchaSiteKey": "5d8c670e-a5f4-4258-ad16-1f42792c9c62", "wgMediaViewerOnClick": true, "wgMediaViewerEnabledByDefault": true, "wgPopupsFlags": 0, "wgVisualEditor": {"pageLanguageCode": "en", "pageLanguageDir": "ltr", "pageVariantFallbacks": "en"}, "wgMFDisplayWikibaseDescriptions": {"search": true, "watchlist": true, "tagline": false, "nearby": true}, "wgWMESchemaEditAttemptStepOversample": false, "wgWMEPageLength": 60000, "wgTestKitchenConfigExperiments": {"overrides": {"enrolled": {"actioned": {"subject": {"id": 1, "wgEditSubmitButtonLabelPublish": true}}}}}}}}

Create a BeautifulSoup object from the HTML response

[17]: # Use BeautifulSoup() to create a BeautifulSoup object from a response text content
      Soup = BeautifulSoup(response.text, 'html.parser')

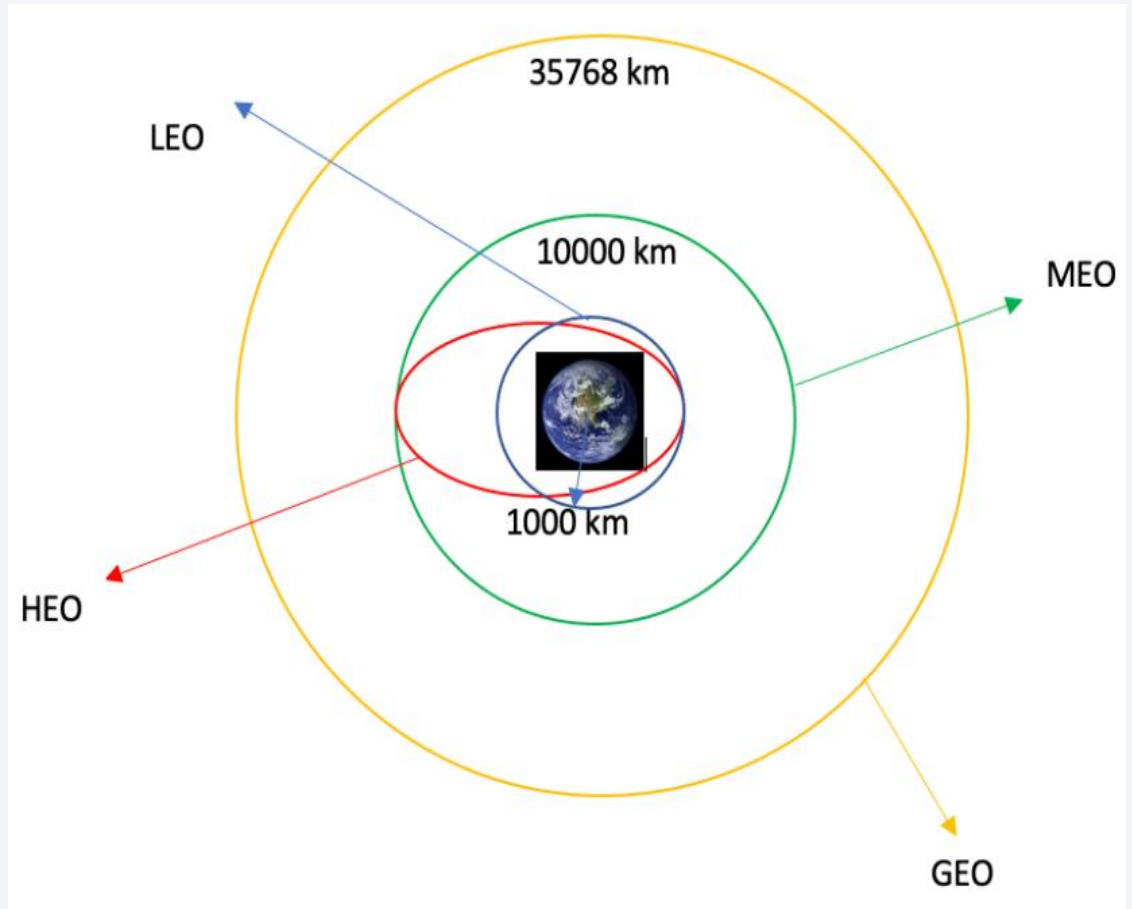
Print the page title to verify if the BeautifulSoup object was created properly

[18]: # Use soup.title attribute
      Soup.title

[18]: <title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

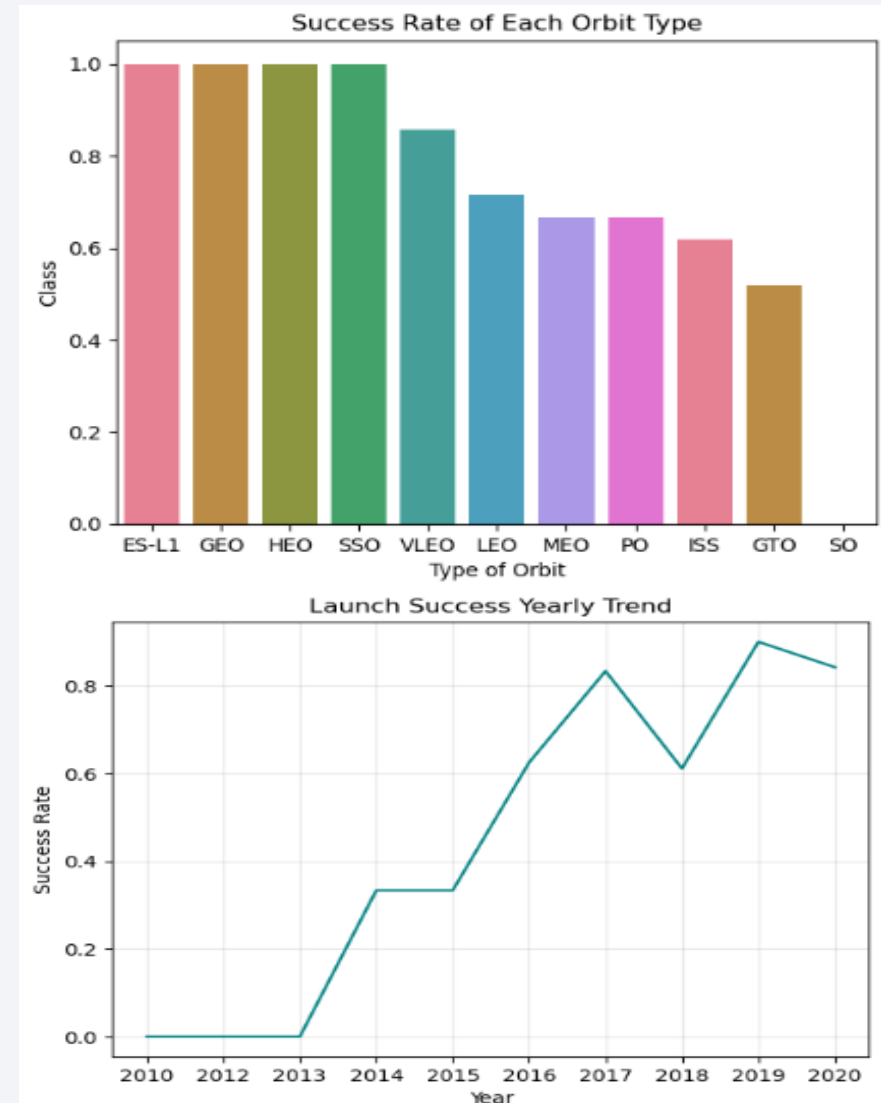
Data Wrangling

- Data analysis was performed to identify launch patterns and determine the training labels.
- We analyzed the number of launches at each launch site and examined the occurrence of different orbit types.
- A landing outcome label was created from the outcome column to classify landing success and failure.
- The processed results were then exported into a CSV file for further analysis and machine learning modeling.
- Notebook Link:
<https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/Data%20Wrangling.ipynb>



EDA with Data Visualization

- We explored the dataset by visualizing the relationship between flight number and launch site.
- Additional analysis was performed to examine the relationship between payload mass and launch site.
- We analyzed the success rate associated with different orbit types to identify performance trends.
- Visualizations were also created to study the relationship between flight number and orbit type.
- Furthermore, we examined the yearly launch success trend to understand improvements in mission reliability over time.
- The link to the notebook:
<https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/EDA%20with%20Data%20Visualization.ipynb>



EDA with SQL

- Exploratory Data Analysis (EDA) was performed using SQL queries to generate insights from the launch dataset.
- Queries were written to identify the unique launch sites involved in SpaceX missions.
- We analyzed the total payload mass carried by NASA (CRS) booster missions.
- The average payload mass carried by the Falcon 9 booster version F9 v1.1 was also calculated.
- SQL analysis was used to determine the total number of successful and failed mission outcomes.
- In addition, failed landing outcomes on drone ships were analyzed along with their booster versions and launch site names.
- The link to the notebook is: <https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/EDA%20Using%20SQL.ipynb>

Build an Interactive Map with Folium

- Interactive visual analytics were performed using Folium maps to analyze SpaceX launch sites and mission outcomes.
- Launch sites were marked on the map using markers, circles, and lines to represent launch success and failure for each location.
- The launch outcome feature was converted into binary classes, where 0 represented failure and 1 represented success.
- Color-coded marker clusters were used to identify launch sites with relatively higher success rates.
- Distance analysis was also conducted to measure the proximity of launch sites to nearby geographical features.
- We analyzed whether launch sites were located near railways, highways, and coastlines.
- Additionally, we examined whether launch sites maintained a certain distance away from nearby cities and populated areas.
- The link to the Notebook: <https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/Interactive%20Launch%20Site%20Visualization%20Using%20Folium.ipynb>

Build a Dashboard with Plotly Dash

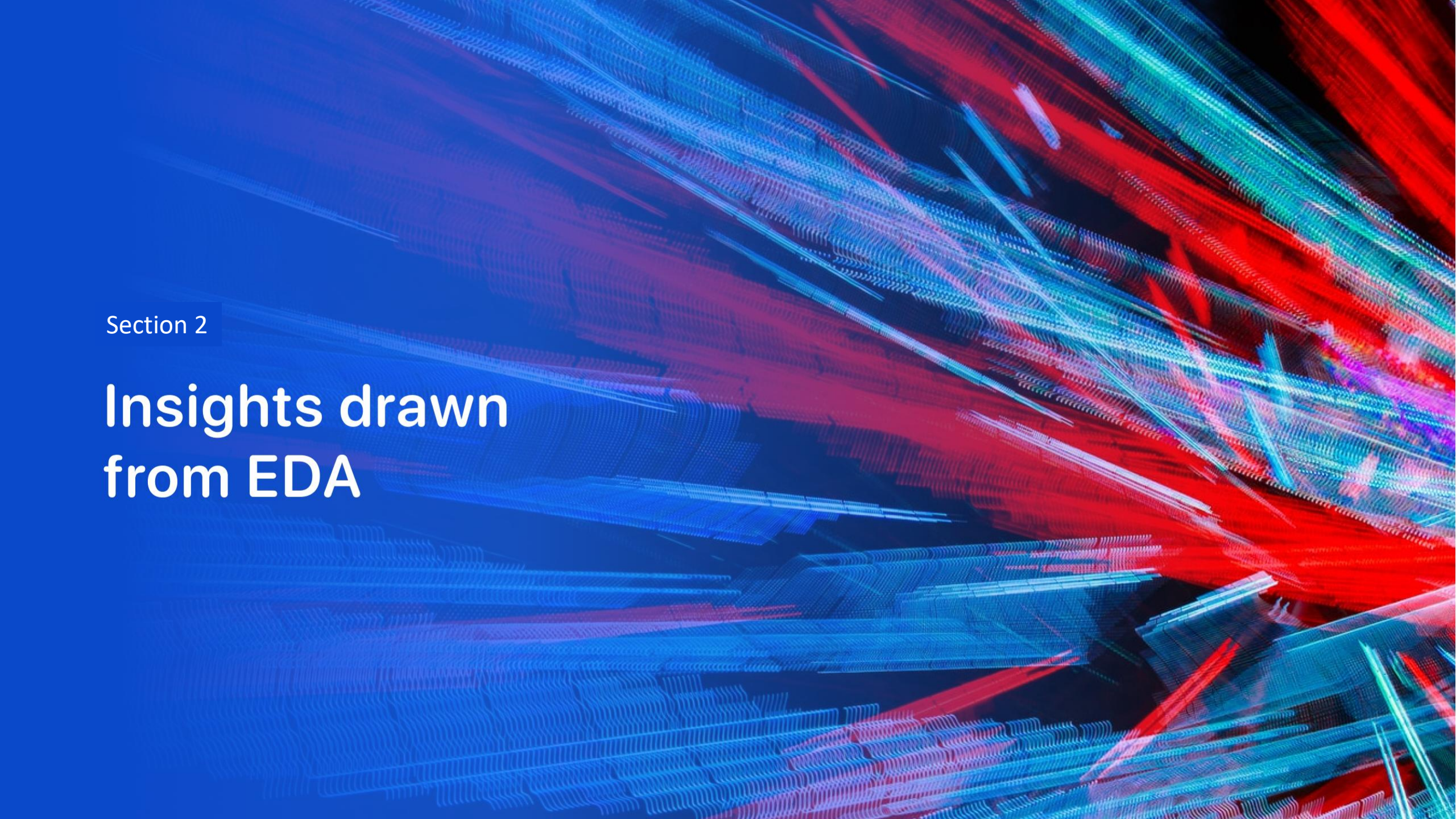
- We developed an interactive dashboard using Plotly Dash to visualize and analyze SpaceX launch data.
- Pie charts were created to display the total number of launches for different launch sites.
- Scatter plots were used to analyze the relationship between payload mass and launch outcomes across different booster versions.
- The dashboard enabled interactive exploration of launch performance, mission success, and payload trends.
- The link to the Interactive Dash File : <https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)

- The dataset was loaded using NumPy and pandas for preprocessing and analysis. Data transformation techniques were applied before splitting the dataset into training and testing sets.
- Multiple machine learning classification models were developed to predict Falcon 9 landing outcomes.
- Hyperparameter tuning was performed using GridSearchCV to optimize model performance.
- Model accuracy was used as the primary evaluation metric during training and testing.
- Finally, the best-performing classification model was identified based on its overall accuracy and predictive capability.
- The link to the notebook : <https://github.com/RiyaBhandari07/IBM-Data-Science-Capstone-Project-SpaceX/blob/main/Machine%20Learning%20Prediction.ipynb>

Results

- Exploratory Data Analysis Visualization Results
- Interactive Analytics Visualization Results
- Machine Learning Prediction Results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is high-tech and digital.

Section 2

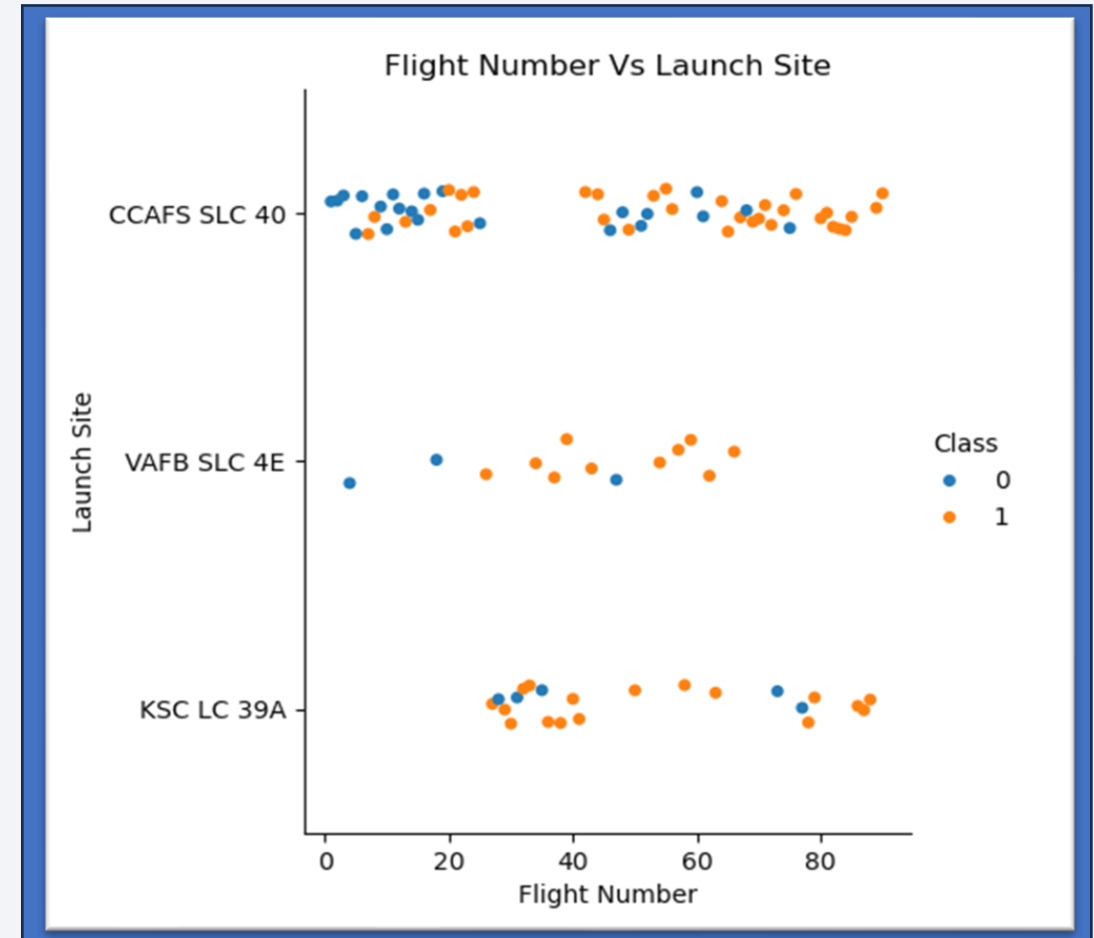
Insights drawn from EDA

Flight Number vs. Launch Site



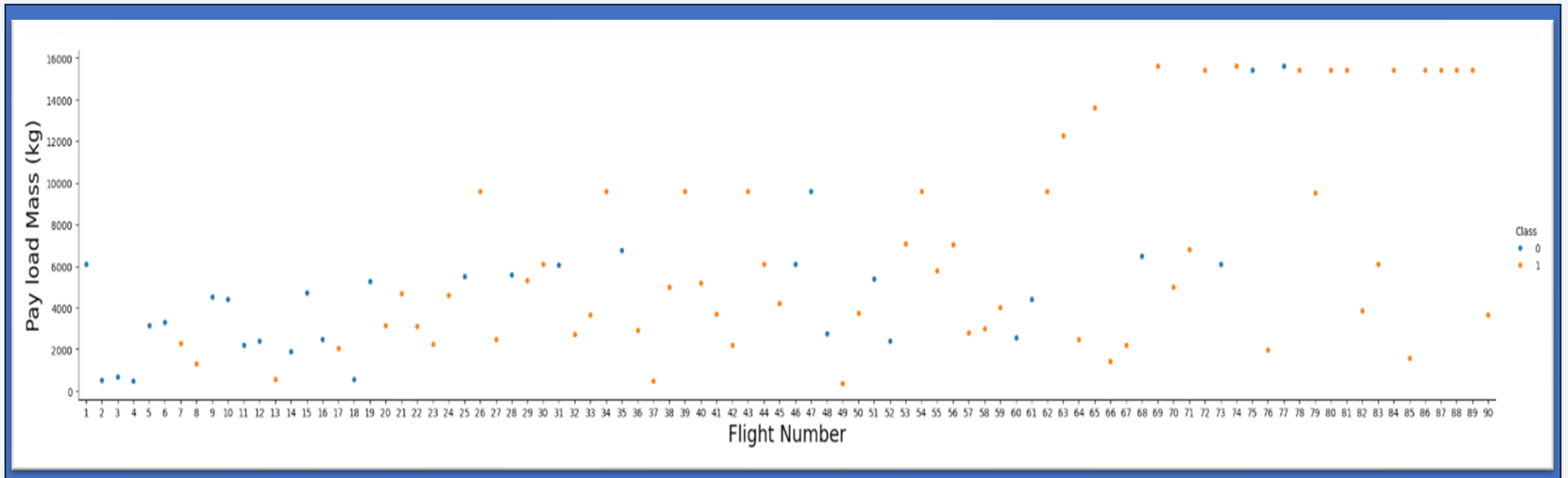
Findings!

- CCAFS SLC 40 recorded the highest number of launches and showed a higher proportion of successful landings as the flight number increased.
- VAFB SLC 4E had comparatively fewer launches, but most of the later missions were successful, indicating improved landing performance over time.
- KSC LC 39A showed a mix of successful and failed launches initially; however, later flight numbers demonstrated more consistent landing success rates.
- Overall, the graph suggests that landing success improved over time across all launch sites, likely due to advancements in booster recovery technology and operational experience.



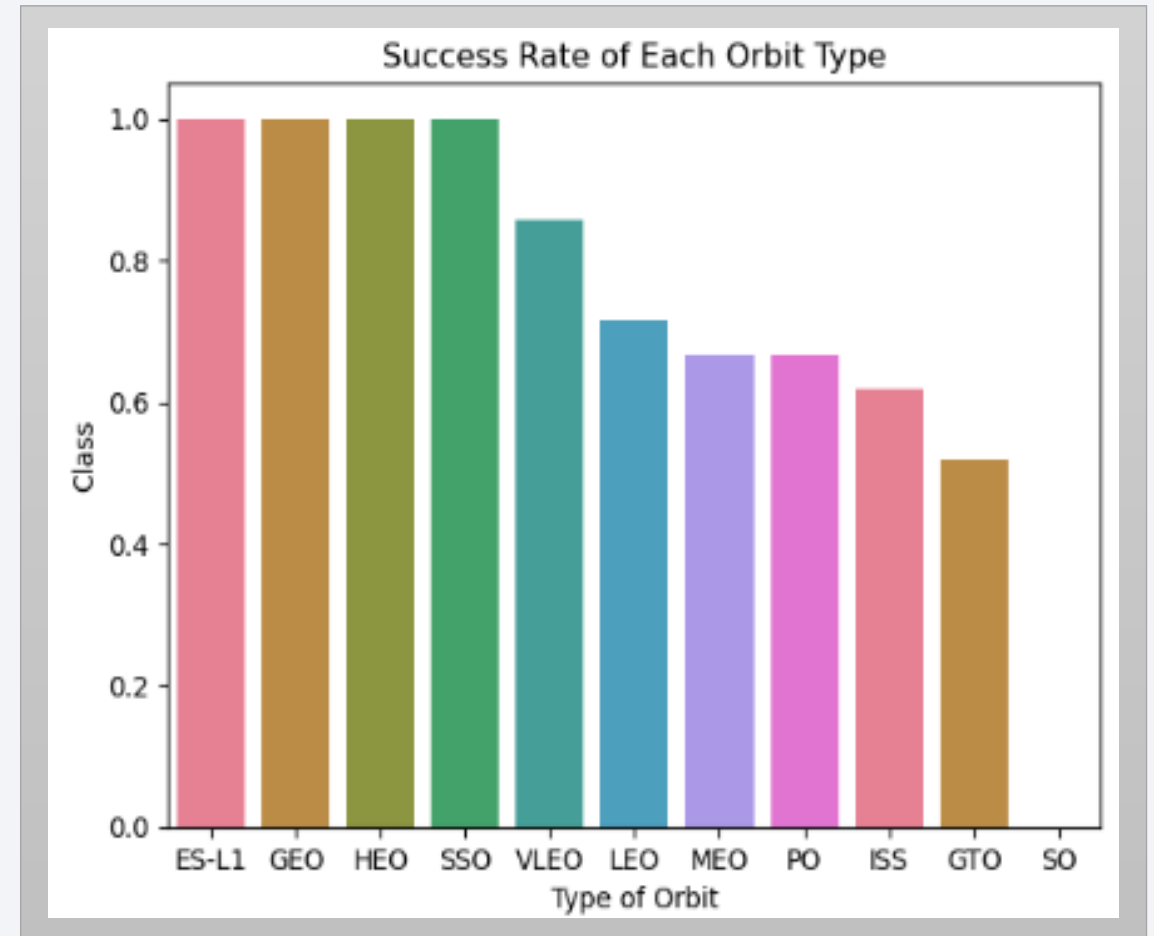
Payload vs. Launch Site

- The graph shows that launch success rates generally improved as the flight number increased, indicating advancements in SpaceX booster recovery technology over time.
- Higher payload masses, especially in later flight numbers, were associated with a greater number of successful landings, demonstrating improved capability in handling heavier payload missions.

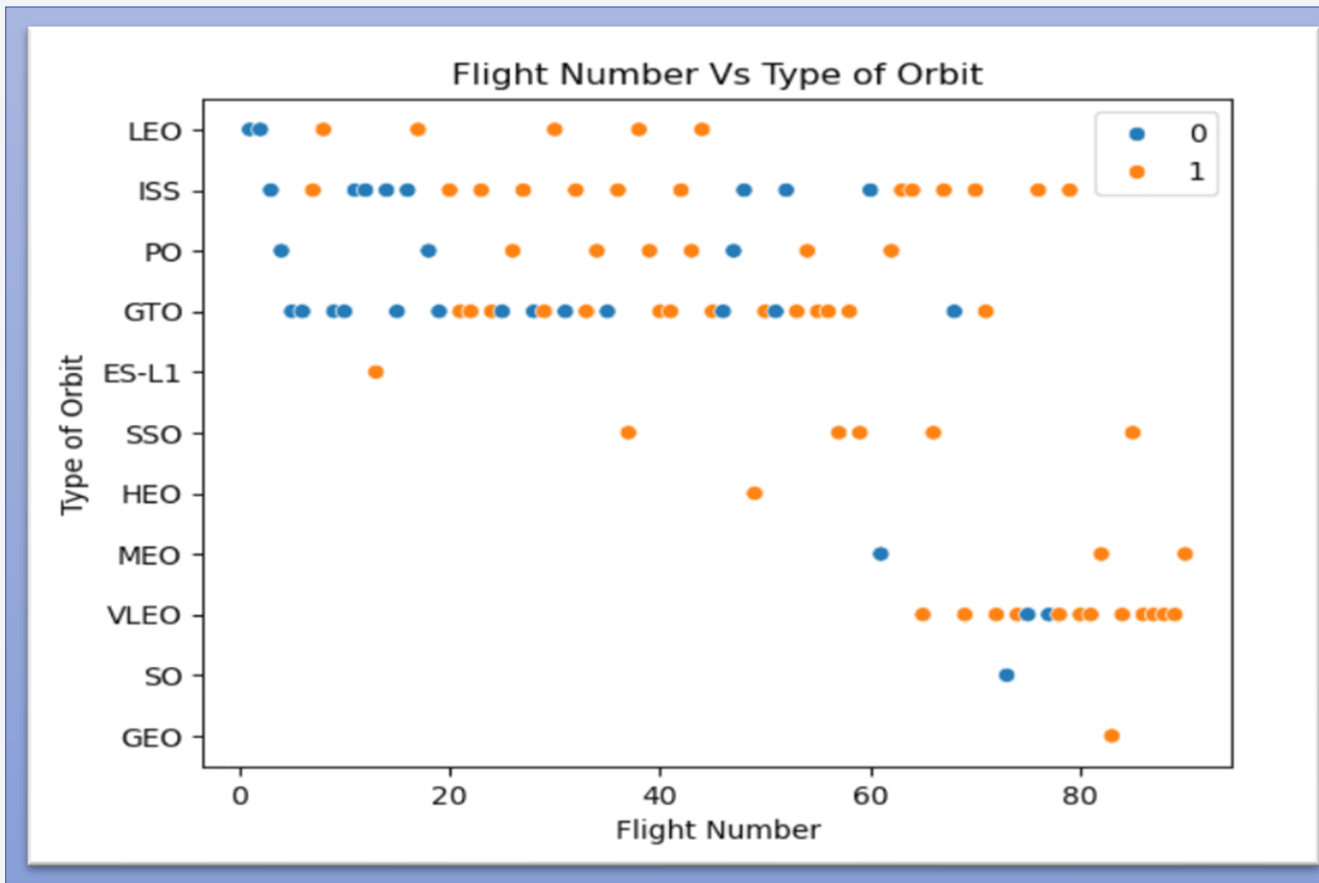


Success Rate vs. Orbit Type

- Orbit types such as ES-L1, GEO, HEO, and SSO achieved nearly 100% landing success rates, indicating highly reliable mission outcomes for these orbits.
- VLEO and LEO missions showed moderately high success rates, suggesting consistent booster recovery performance across frequently used orbit types.
- GTO missions had comparatively lower success rates, while SO missions recorded the lowest success rate, indicating that these orbit types present greater landing and recovery challenges.



Flight Number vs. Orbit Type

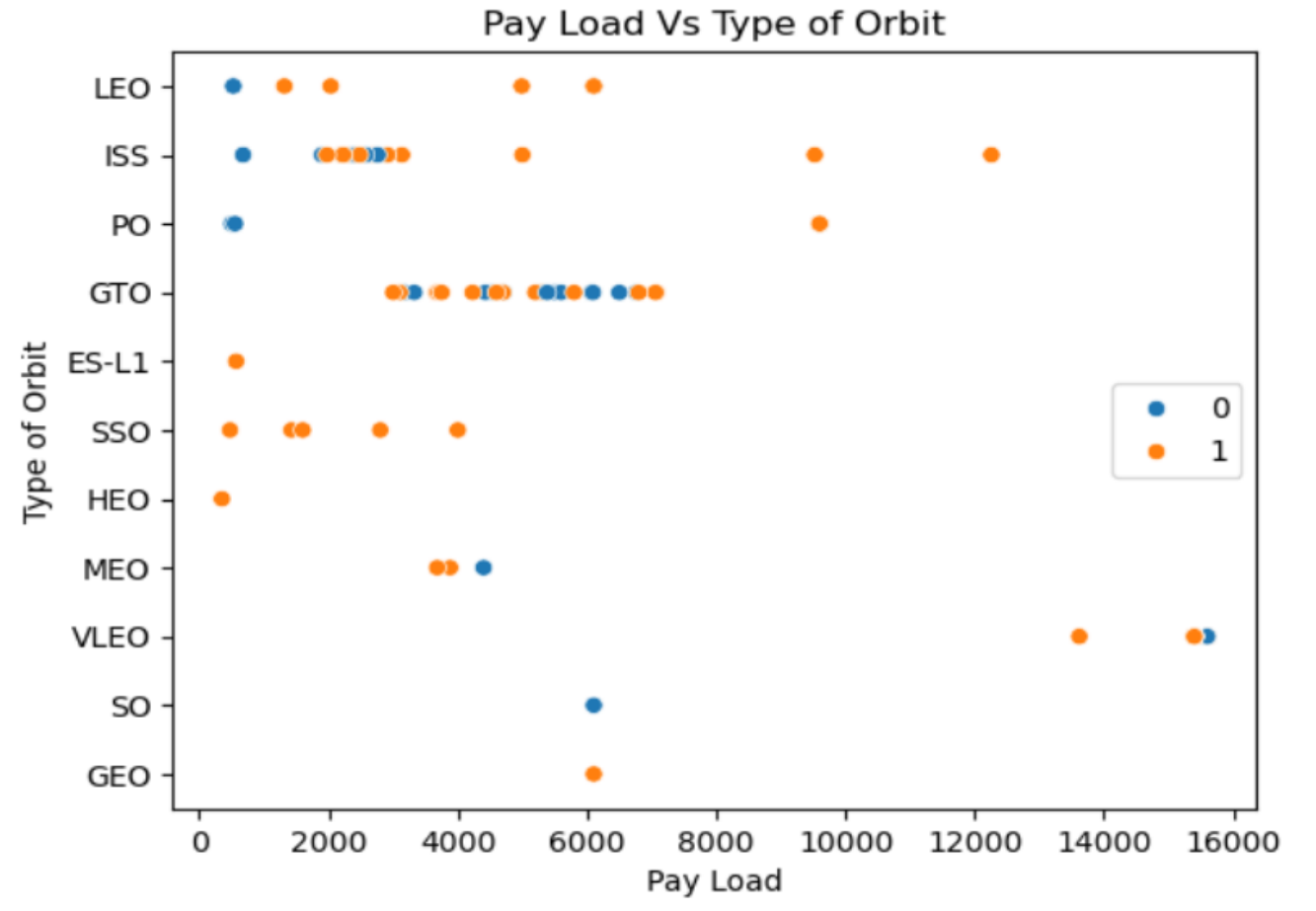


- GTO and ISS missions showed a mix of successful and failed landings in the earlier flight numbers, but landing success became more consistent in later missions.
- Orbit types such as VLEO, SSO, GEO, and HEO showed mostly successful landings, indicating improved booster recovery performance as SpaceX gained more operational experience.

Payload vs. Orbit Type

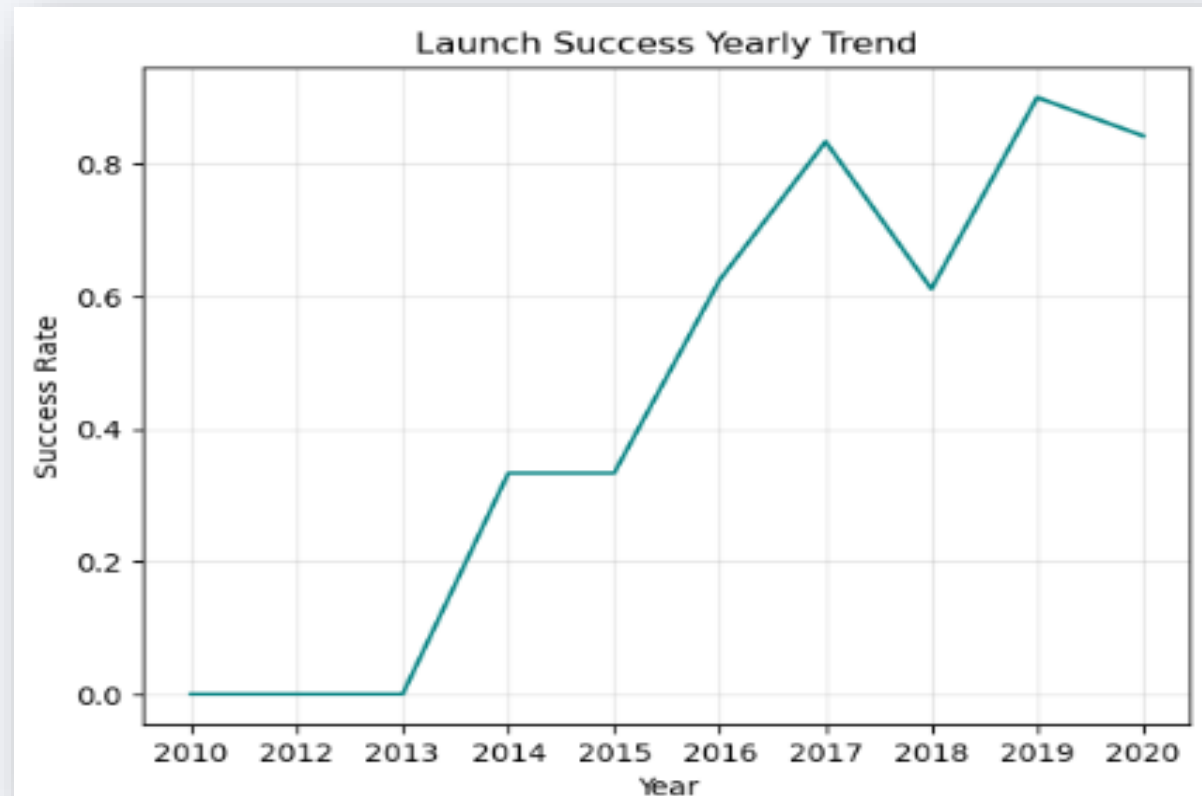


- GTO missions carried a wide range of payload masses and showed a mix of successful and failed landings, indicating greater mission complexity for this orbit type.
- VLEO missions were associated with the highest payload masses and achieved mostly successful landings, demonstrating improved booster recovery capability for heavier payloads.
- Orbit types such as SSO, HEO, and GEO showed predominantly successful landings despite varying payload masses, suggesting reliable mission performance across these orbit categories.



Launch Success Yearly Trend

- The launch success rate showed a significant improvement over the years, indicating advancements in SpaceX landing and recovery technology.
- Although there was a slight decline in success rate in 2018, the overall trend remained positive, reaching the highest success rates during 2019 and 2020.



All Launch Site Names

- We used the SQL keyword DISTINCT in Python to retrieve only the unique launch site names from the SpaceX dataset.

Task 1

Display the names of the unique launch sites in the space mission

```
[10]: %sql SELECT DISTINCT Launch_Site FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[10]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```


Launch Site Names Begin with 'CCA'

- This SQL query retrieves the first five records from the SpaceX dataset where the launch site name begins with “CCA” using the LIKE operator and LIMIT clause.

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[15]: %sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

```
[15]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- This SQL query calculates the total payload mass carried by boosters launched for NASA (CRS) missions using the SUM aggregation function.
- The Total Payload Mass for NASA (CRS) is 45596

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[9]: %sql SELECT Customer, SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass FROM SPACEXTBL WHERE Customer LIKE 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[9]:
```

Customer	Total_Payload_Mass
----------	--------------------

NASA (CRS)	45596
------------	-------

Average Payload Mass by F9 v1.1

- This SQL query calculates the average payload mass carried by the Falcon 9 booster version F9 v1.1 using the AVG aggregation function.
- The Average Payload for booster version F9 v1.1 is 2928.4

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[39]: %sql SELECT Booster_Version, AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

Done.

```
[39]: Booster_Version Average_Payload_Mass
```

F9 v1.1	2928.4
---------	--------

First Successful Ground Landing Date

- This SQL query uses the MIN() function to find the earliest date on which a successful ground pad landing was achieved in the SpaceX dataset.
- Output: The first successful ground pad landing was achieved on 2015-12-22.

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[43]: %sql SELECT Landing_Outcome, MIN(Date) AS Date FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';  
* sqlite:///my_data1.db  
Done.
```

```
[43]:
```

Landing_Outcome	Date
Success (ground pad)	2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- This SQL query retrieves the booster versions that achieved successful drone ship landings with payload masses between 4000 kg and 6000 kg using the BETWEEN operator.
- Output: The booster versions identified were F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2.

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[44]: %%sql SELECT Booster_Version FROM SPACEXTBL
      WHERE Landing_Outcome = 'Success (drone ship)' AND
      PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
* sqlite:///my_data1.db
```

Done.

```
[44]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Total Number of Successful and Failure Mission Outcomes

- This SQL query uses conditional counting with CASE statements to calculate the total number of successful and failed mission outcomes in the SpaceX dataset.
- Output: The dataset contains 100 successful mission outcomes and 1 failed mission outcome.

```
Task 7
List the total number of successful and failure mission outcomes

[15]: %%sql SELECT
      COUNT(CASE
        WHEN Mission_Outcome LIKE 'Success%'
        THEN 1
      END) as Successful,
      COUNT(CASE
        WHEN Mission_Outcome LIKE 'Failure%'
        THEN 1
      END) as Failure
      FROM SPACEXTBL;

* sqlite:///my_data1.db
Done.
[15]:
```

Successful	Failure
100	1

Boosters Carried Maximum Payload

- This SQL query uses a subquery with the **MAX()** aggregation function to identify all booster versions that carried the maximum payload mass in the SpaceX dataset.

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
[49]: %%sql SELECT Booster_Version FROM SPACEXTBL
      WHERE PAYLOAD_MASS_KG_ = (
        SELECT MAX(PAYLOAD_MASS_KG_) FROM SPACEXTBL) ;
```

```
* sqlite:///my_data1.db
```

Done.

```
[49]: Booster_Version
```

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- This SQL query retrieves the months, booster versions, and launch sites for failed drone ship landing outcomes that occurred during the year 2015 using the **strftime()** function for date filtering.
- **Output:** In 2015, failed drone ship landings occurred in January and April involving booster versions F9 v1.1 B1012 and F9 v1.1 B1015 at the CCAFS LC-40 launch site.

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

[50]: %%sql

```
SELECT
    strftime('%m', Date) AS Month,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTBL
WHERE Landing_Outcome LIKE 'Failure%drone ship%'
AND strftime('%Y', Date) = '2015';
```

* sqlite:///my_data1.db

Done.

[50]:

Month	Landing_Outcome	Booster_Version	Launch_Site
-------	-----------------	-----------------	-------------

01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
----	----------------------	---------------	-------------

04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
----	----------------------	---------------	-------------

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- This SQL query counts and ranks different landing outcomes between 2010-06-04 and 2017-03-20 in descending order using the **COUNT()** function, **GROUP BY** clause, and **ORDER BY** statement.

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[51]: %%sql
SELECT
    Landing_Outcome,
    COUNT(*) AS Outcome_Count
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

* sqlite:///my_data1.db
Done.

```
[51]:
```

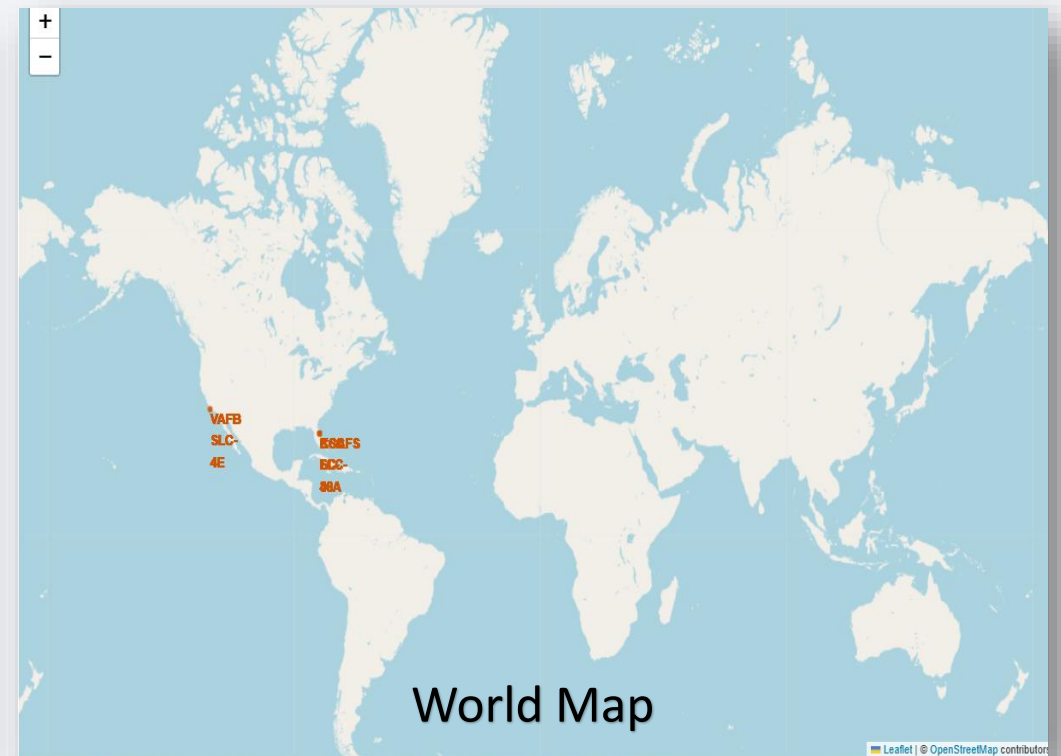
Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

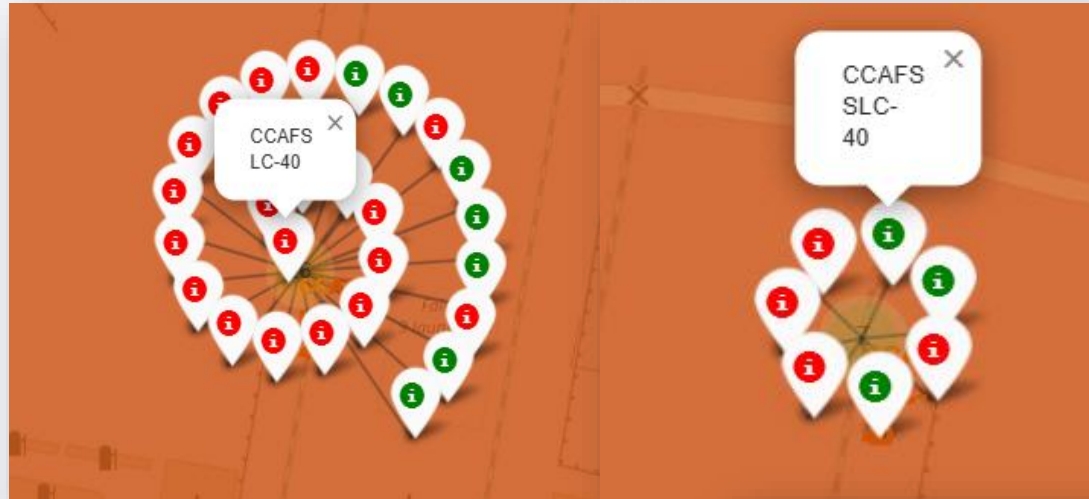
Launch Sites Proximities Analysis

All launch sites global map markers



SpaceX launch Sites are all in USA along the coastline of California and Florida

Launch Site Visualization with Color-Coded Markers

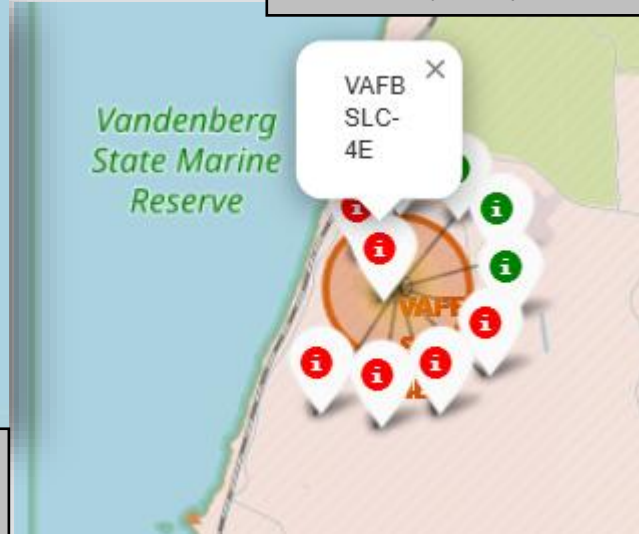


- Green markers indicate successful landings
- Red markers represent failed landings.

VAFB SLC-4E is located at Vandenberg Space Force Base, California, USA/

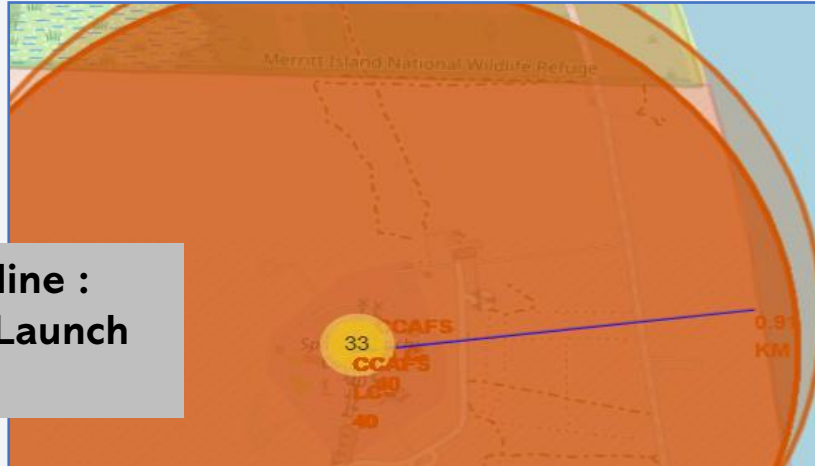


- CCAFS LC-40 is located at Cape Canaveral Space Force Station, Florida, USA.
- KSC LC-39A is located at Kennedy Space Center, Florida, USA.

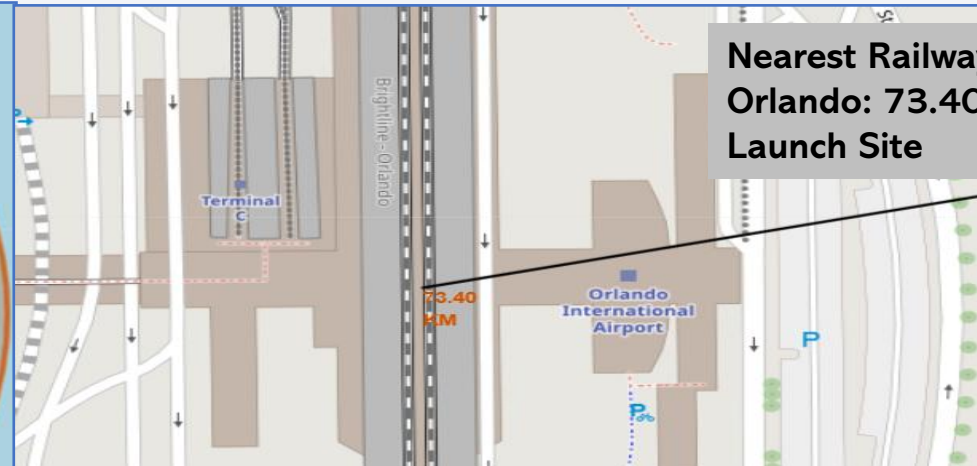


Launch Site Distance to Landmarks

**Nearest Coastline :
0.90km from Launch
Site**



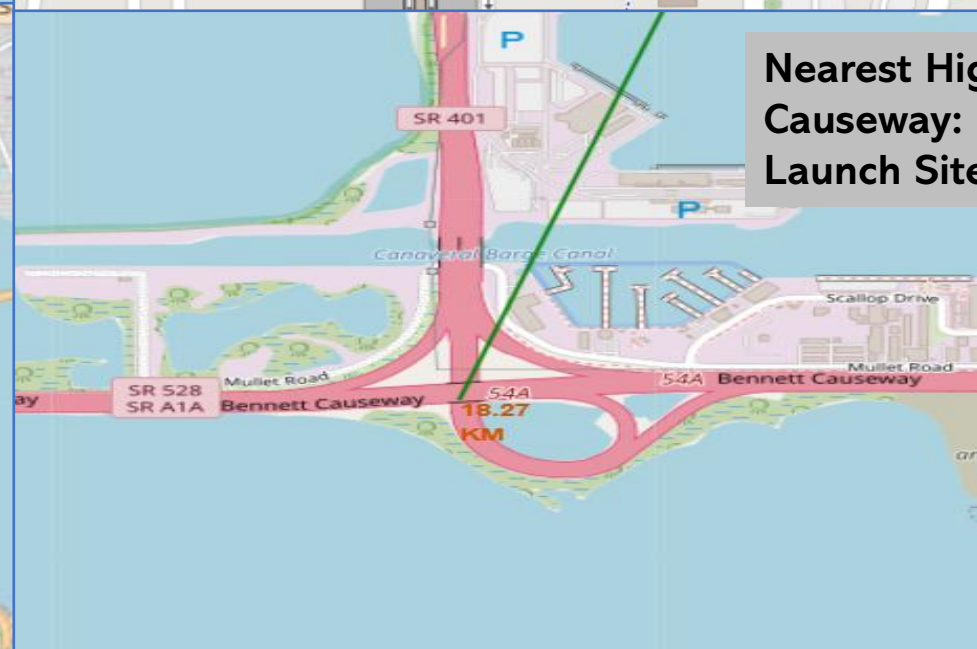
**Nearest Railway Station in
Orlando: 73.40km from
Launch Site**



**Nearest City Melbourne:
54.53km from Launch
Site**



**Nearest Highway Bennett
Causeway: 18.27km from
Launch Site**



Launch Site Proximity Concerns

Are launch sites in close proximity to coastline?

- The launch pad is extremely close to the coastline, at a distance of only about **0.91 km**. This is a strategic advantage because rockets are generally launched over the ocean to reduce risks to populated areas in case of launch failure or falling debris. Coastal locations also provide clear launch trajectories and improve overall launch safety.

Are launch sites in close proximity to highways?

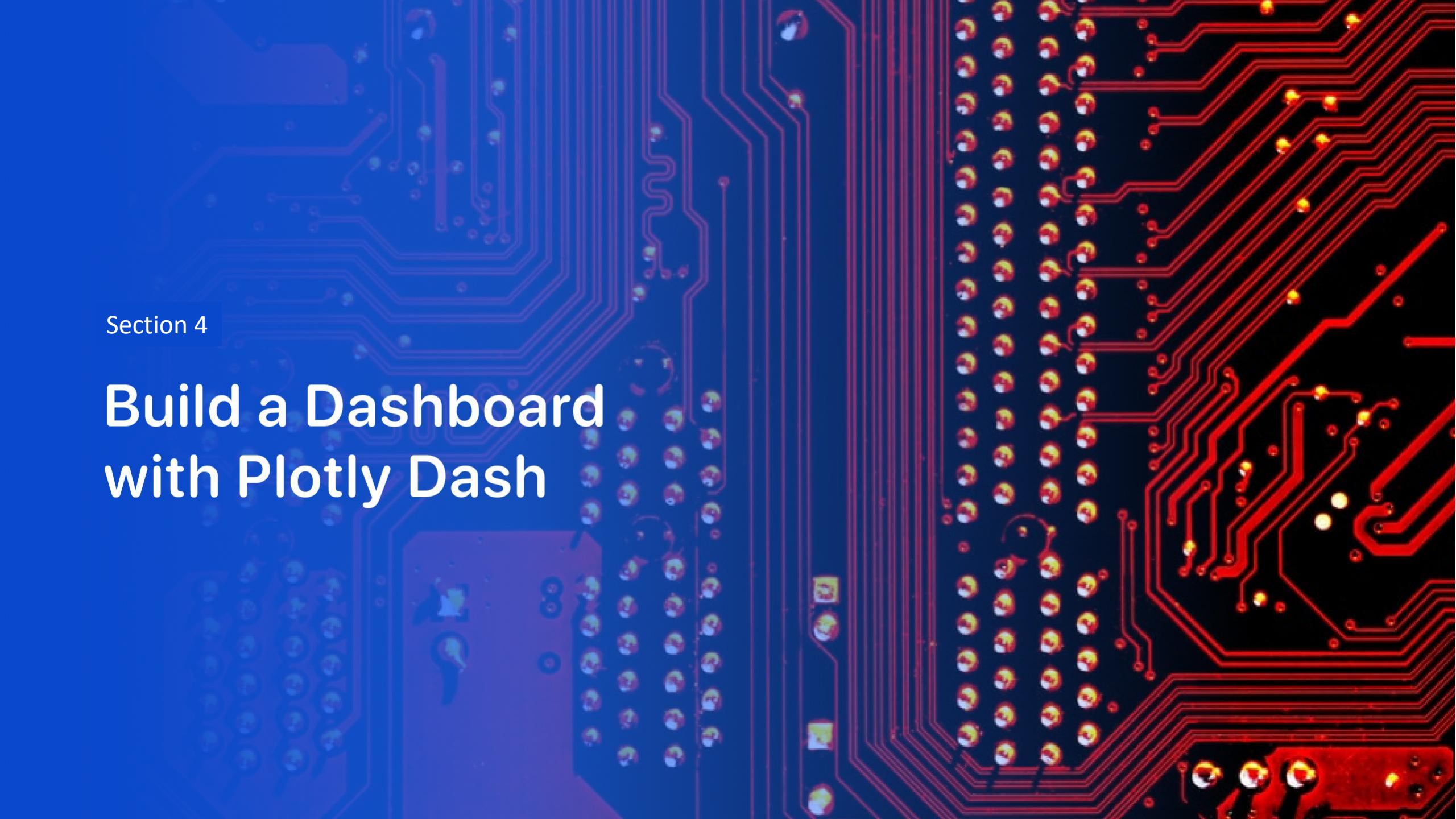
- The launch site is also located very close to a highway, approximately **18.27 km** away (Bennett Causeway). This close highway access is important for transportation and logistics, allowing easy movement of rocket components, fuel systems, equipment, and personnel to the launch facility.

Are launch sites in close proximity to railways?

- The nearest railway station is located in **Orlando** around **73.40 km** from the launch pad and is positioned near two other launch sites. Rail connectivity can support the transport of heavy aerospace equipment and large structural components that may be difficult to move entirely by road.

Do launch sites keep certain distance away from cities?

- The nearest city, **Melbourne** is approximately **54.53 km** away from the launch site. This indicates that the launch facility maintains a safe distance from populated residential areas. Such separation helps minimize risks associated with rocket launches, including explosions, noise, vibrations, and hazardous debris.

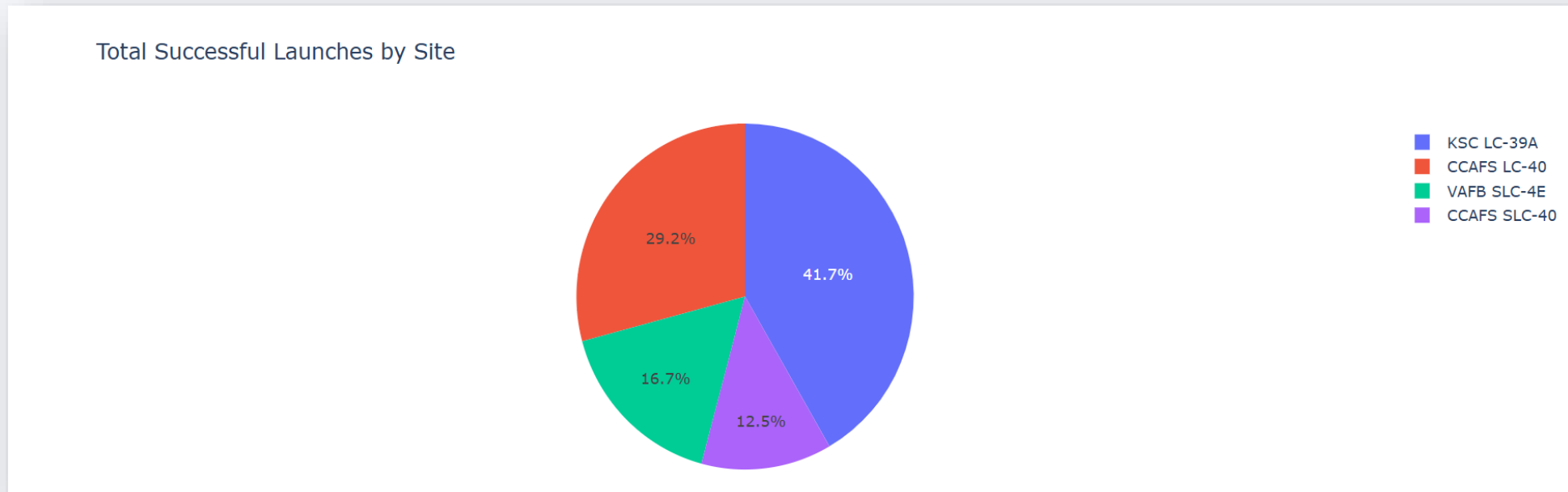


Section 4

Build a Dashboard with Plotly Dash

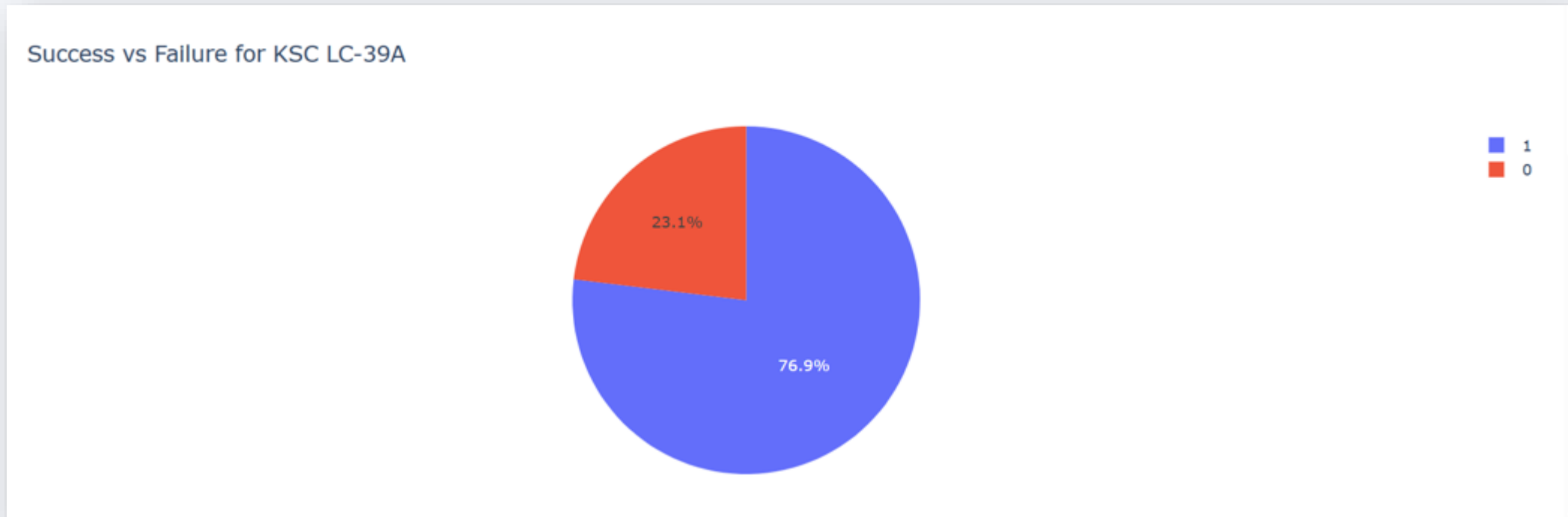
Pie chart showing the success percentage achieved by each launch site

The pie chart shows that KSC LC-39A recorded the highest proportion of successful launches, followed by CCAFS LC-40, while VAFB SLC-4E and CCAFS SLC-40 contributed comparatively smaller shares of successful missions.

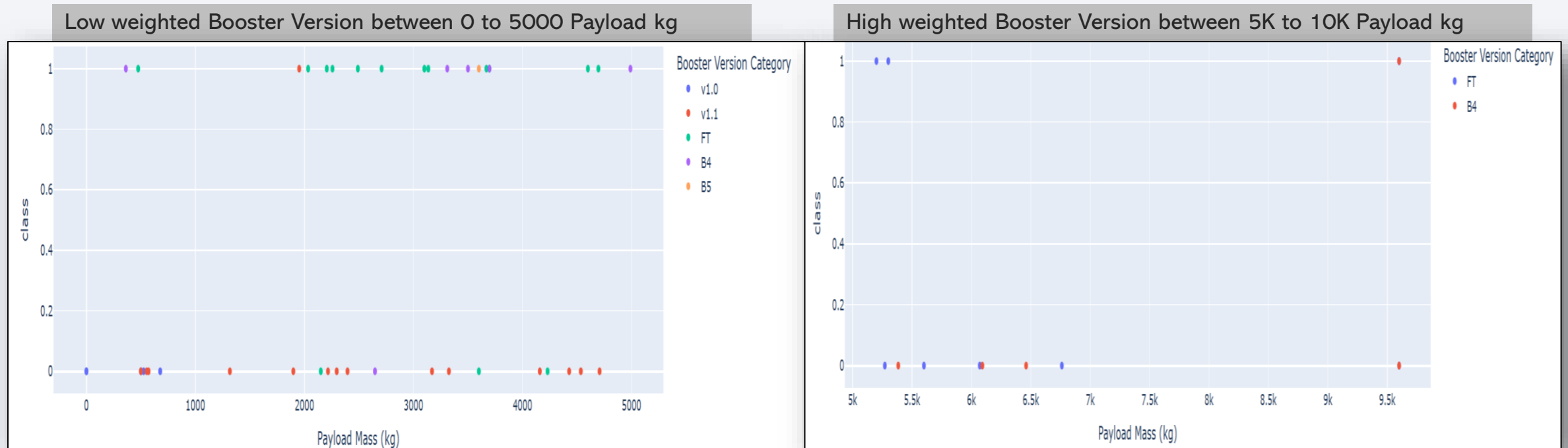


Pie Chart Representing the Launch Site with the Highest Success Rate

The pie chart indicates that KSC LC-39A achieved a high launch success rate, with approximately 76.9% successful landings and 23.1% failed landings.



Scatter Plot Showing Payload Mass vs Launch Outcome Across All Launch Sites with Interactive Payload Range Selection



The scatter plot illustrates the relationship between payload mass and launch outcomes for different booster versions, showing that newer booster categories achieved more successful landings across a wider payload range.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

The Decision Tree classifier achieved the highest classification accuracy among all the machine learning models evaluated.

TASK 12

Find the method performs best:

```
In [30]: models = {'KNeighbors' : knn_cv.best_score_,
                  'DecisionTree': tree_cv.best_score_,
                  'LogisticRegression': logreg_cv.best_score_,
                  'SupportVectorMachince': svm_cv.best_score_}

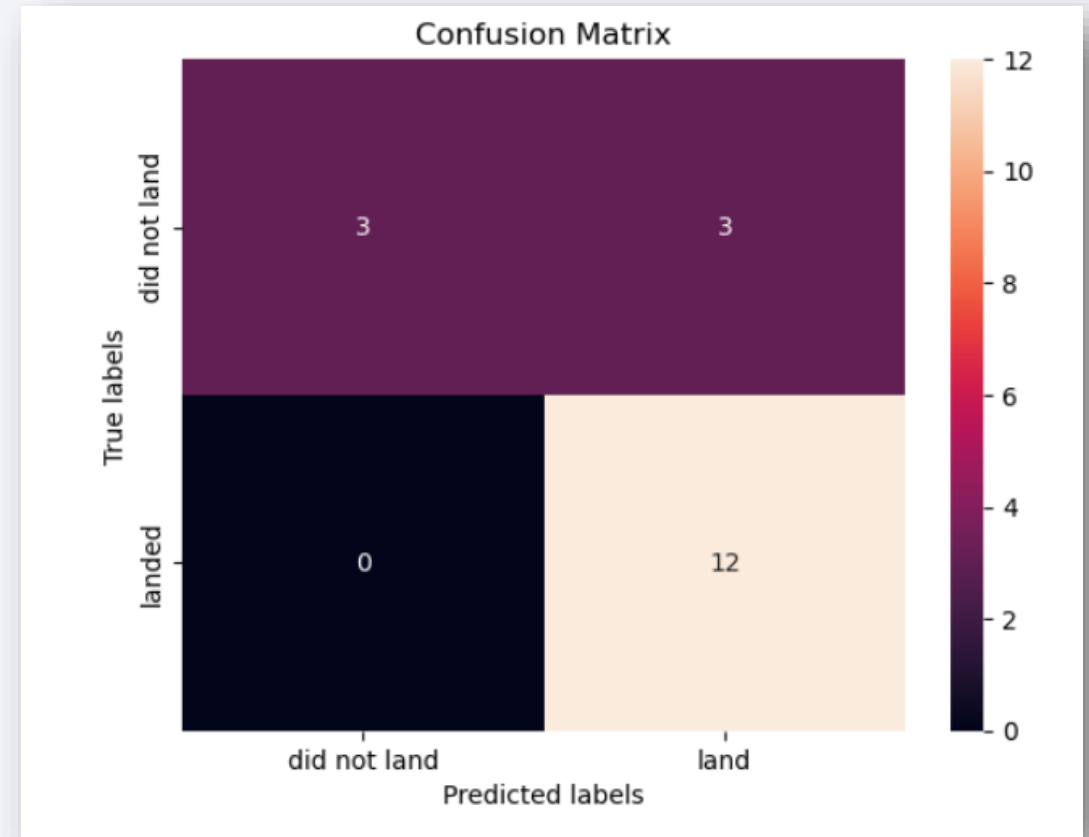
best_algorithm = max(models, key=models.get)
print('Best model is', best_algorithm, 'with a score of', models[best_algorithm])
if best_algorithm == 'DecisionTree':
    print("Best parameters are:", tree_cv.best_params_)
if best_algorithm == 'KNeighbors':
    print("Best parameters are:", knn_cv.best_params_)
if best_algorithm == 'LogisticRegression':
    print("Best parameters are:", logreg_cv.best_params_)
if best_algorithm == 'SupportVectorMachince':
    print("Best parameters are:", svm_cv.best_params_)
```

Best model is DecisionTree with a score of 0.8767857142857143

Best parameters are: {'criterion': 'entropy', 'max_depth': 4, 'max_features': 'log2', 'min_samples_leaf': 2, 'min_samples_split': 10, 'splitter': 'random'}

Confusion Matrix

- The Decision Tree model correctly predicted 12 successful landings and 3 unsuccessful landings, showing strong overall classification performance.
- The model produced no false negatives, meaning all actual successful landings were correctly identified as successful.
- However, the model misclassified 3 failed landings as successful landings, indicating some difficulty in distinguishing failure cases.



Conclusions

- The sites with a higher number of launches generally achieved greater landing success rates, indicating improved operational experience over time.
- Falcon 9 launch success rates steadily improved from 2013 to 2020 due to advancements in reusable rocket technology and landing systems.
- Orbit types such as ES-L1, GEO, HEO, SSO, and VLEO demonstrated the highest landing success rates, highlighting strong mission reliability across these orbital categories.
- Among all launch sites, KSC LC-39A recorded the highest number of successful launches, making it the most successful launch site
- The Decision Tree classifier achieved the best performance and provided the highest accuracy for predicting Falcon 9 landing outcomes.

Thank you!

